Geofinitism and the Finite Tractus: Principia Geometrica

A Trajectory Toward Finite Symbolic Mechanics

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Geofinitism: A Continuing Path

It seems to me there's a tendency, particularly in mathematics and the sciences, to regard large foundational works as definitive statements. A system is constructed, formalised, and presented as a completed object that can then be studied, critiqued, and extended. Historically, such works have often been positioned as endpoints: Euclid's *Elements*, Newton's *Principia*, or the formal developments of the twentieth century.

Yet in practice, the creation of such works rarely follows this neat pattern. They do not emerge fully formed, nor do they mark the conclusion of a line of thought. Rather, they are situated within an ongoing trajectory: shaped by prior ideas, constrained by the concepts available at the time of writing, and inevitably reinterpreted as new structures emerge. It is in this spirit that my earlier work, the *Finite Tractus: Principia Geometrica,* should be understood.

The Emergence of a Large-Scale Work

The document arose during a period of intense exploration across several domains: finite mathematics, physical modelling, symbolic systems, and the behaviour of language within computational frameworks. At the time, these threads were far from unified. They were connected through insight and repeated reconstruction, but the underlying structure that would later bring clarity to their relationships had not yet been fully articulated.

The work itself reflects this trajectory. It attempts a large-scale synthesis, bringing together ideas of finiteness, measurement, geometry, and symbolic interaction into a single framework. In doing so, it necessarily reaches beyond the stabilised concepts available at the time. It explores, tests, and constructs, often simultaneously.

As a result, the document carries a provenance of uncertainty. It contains stable insights, local structures that hold under reconstruction, and also regions that reflect the limits of the conceptual tools then available. This is the nature of trying to build a new foundational philosophy emerging from a foundational commitment to finite measurements and all that involves.

What Was Not Yet Visible

Since the initial development of the document, several clarifying structures have continued to emerge. In particular, the concepts of **commitment**, **admissibility**, and **consensus.** These concepts provide a more precise way of understanding how symbolic systems stabilise and how Geofinitism may or may not fit into the wider landscape and achieve any degree of consensus.

At the time of writing, the ideas of commitment, admissibility and consensus were present implicitly, but not yet named or formalised. Therefore the Finite Tractus: Principia Geometrica operates within them, but does not yet describe them. It attempts to stabilise a system without fully articulating the conditions under which that stabilisation occurs.

Similarly, the development of language as a **nonlinear dynamical system**, and the recognition of symbolic meaning as trajectory-dependent within a geometric space, provide a deeper interpretative layer that was not explicitly available during the original construction.

Perhaps, viewed through these later developments, the document becomes clearer; not as a finished structure, but as a generative one. It can be read as an object that both expresses and precedes the framework required to fully situate the work within the wider basin of Geofinitism.

Reframing the Principia

From the perspective of Geofinitism as a developing philosophy it is perhaps useful to consider the status of *Finite Tractus: Principia Geometrica:* it is best considered as a **foundational trajectory**.

In practice, it represents an attempt to stabilise a symbolic system under conditions that were themselves still evolving. For me, it captures a phase in the development of the broader framework. A framework in which many of the core ideas are emerging, but not yet fully resolved into their later forms.

In this sense, the document functions as an attractor within the wider body of work. It gathers together a range of ideas into a coherent, though not final, structure. It provides a space within which these ideas can be reconstructed, tested, and extended.

The value of such a document may not lie in its completion, but in its **coherence and accessibility**. Especially if it is sufficiently stable to be entered, and sufficiently open to support further development.

Placement Within Geofinitism

Releasing the document in its current form reflects this understanding. It is therefore made available not as a finished system, but as part of a broader corpus that includes:

- ongoing Substack articles, which explore narrative and conceptual trajectories
- GitHub repositories, which provide formal structures and working documents
- a dedicated website, which serves as a stable reference point

Together, these elements form a distributed and wider system of symbolic exploration. The *Principia Geometrica* occupies a foundational particular position within this system, alongside the Finite Tractus: The Hidden Geometry of Language and Thought. It is a large, coherent artefact that anchors many of the ideas developed elsewhere.

Importantly, subsequent work does not replace it, but rather reframes and extends it. New documents operate alongside it, drawing upon its structures while hopefully introducing additional layers of clarity and formalisation.

On Stability and Incompleteness

For me and perhaps many writers, there is a natural inclination to seek completion in foundational work. To refine, correct, and unify until a single, stable system emerges. Yet the analysis developed in *Finite Symbolic Mechanics* suggests a different perspective. That is to say, symbolic systems do not achieve stability through completion, but through **reconstruction across multiple contexts**.

From this viewpoint, incompleteness is a condition that enables further exploration. For example, if a system is entirely closed it admits no extension. However, a system that remains coherent and open can continue to evolve.

The *Principia Geometrica* should therefore be read in this light not as an incomplete construction, but as a **measurable symbolic object with internal stability and external**

openness. It marks a boundary within the trajectory of the work: a point at which a system became sufficiently formed to be shared, while still capable of transformation.

Onwards

This document can be dipped into ot used as a reference. Releasing it reflects a shift in my own perspective: from the pursuit of completed systems to the cultivation of stable, finite symbolic trajectories. In this way, I feel my work is now available in the public space as a reference point as the broader system continues to evolve. Below is a list of the contents with a link to the full document.

[*Finite Tractus: Principia Geometrica*]

The document is structured as follows:

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4 The Finite Irreversibility Theorem (FIT)

5 The Manifold of Mathematics

6 Formalizing the Five Pillars

7 A Geofinite Information Theory

8 The Geometry of Pi

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27 The Fractal Borders

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30 The Instability of Symbolic Grounding

31 Distorting the Semantic Manifold

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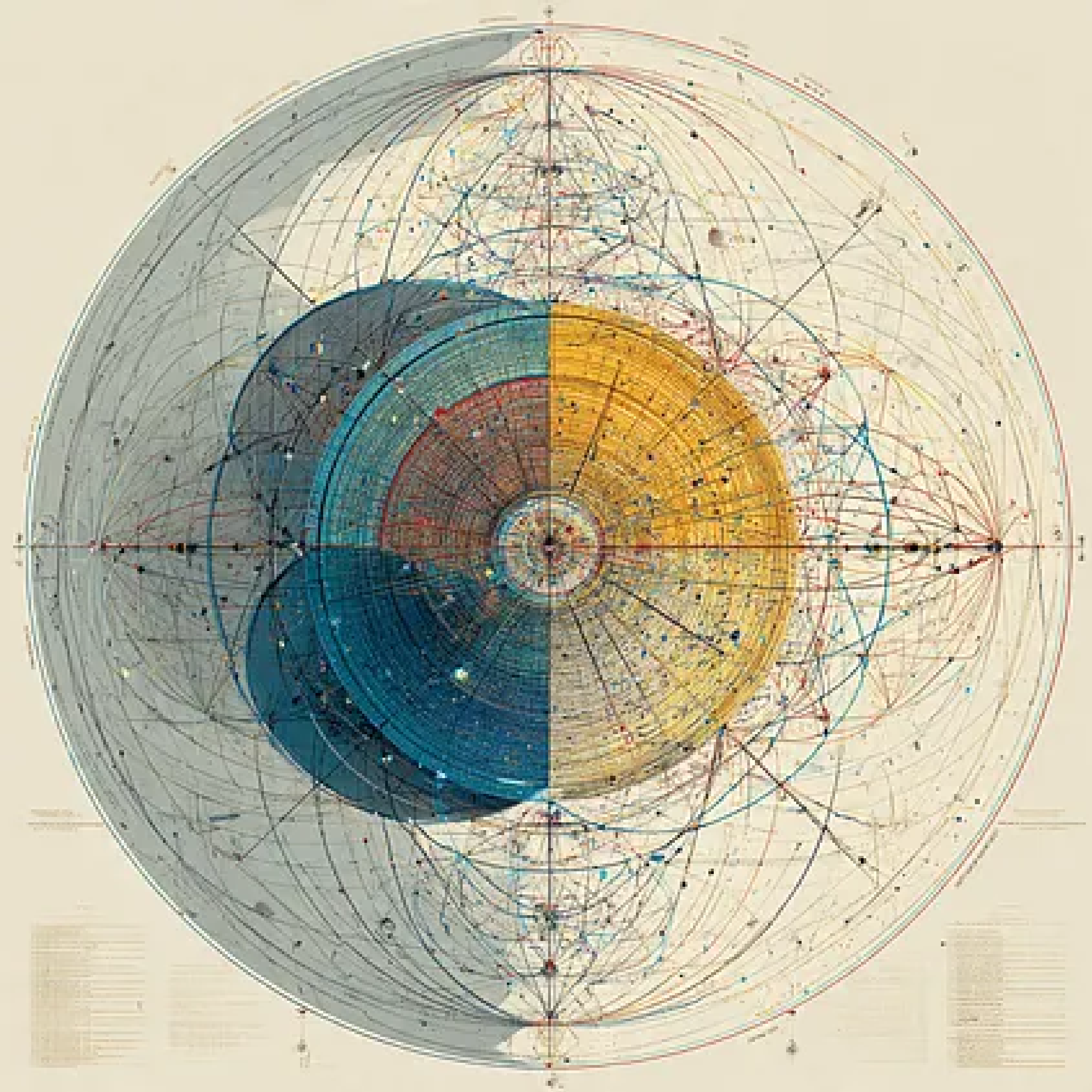
32 Adnotatio I — The Finite Poincaré Conjecture

33 Adnotatio II — The Spherical Attractor of Superposed Word–Forms in a Finite Manifold

34 Adnotatio III — The Finite Asymmetry of Computation: P | NP



~Commitment, Consensus, and Admissibility: Reframing the Past, Language, and Scientific Change

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Introduction

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Geofinitism: Language as a Nonlinear Dynamical System - Attractors, Basins, and the Geometry of Understanding

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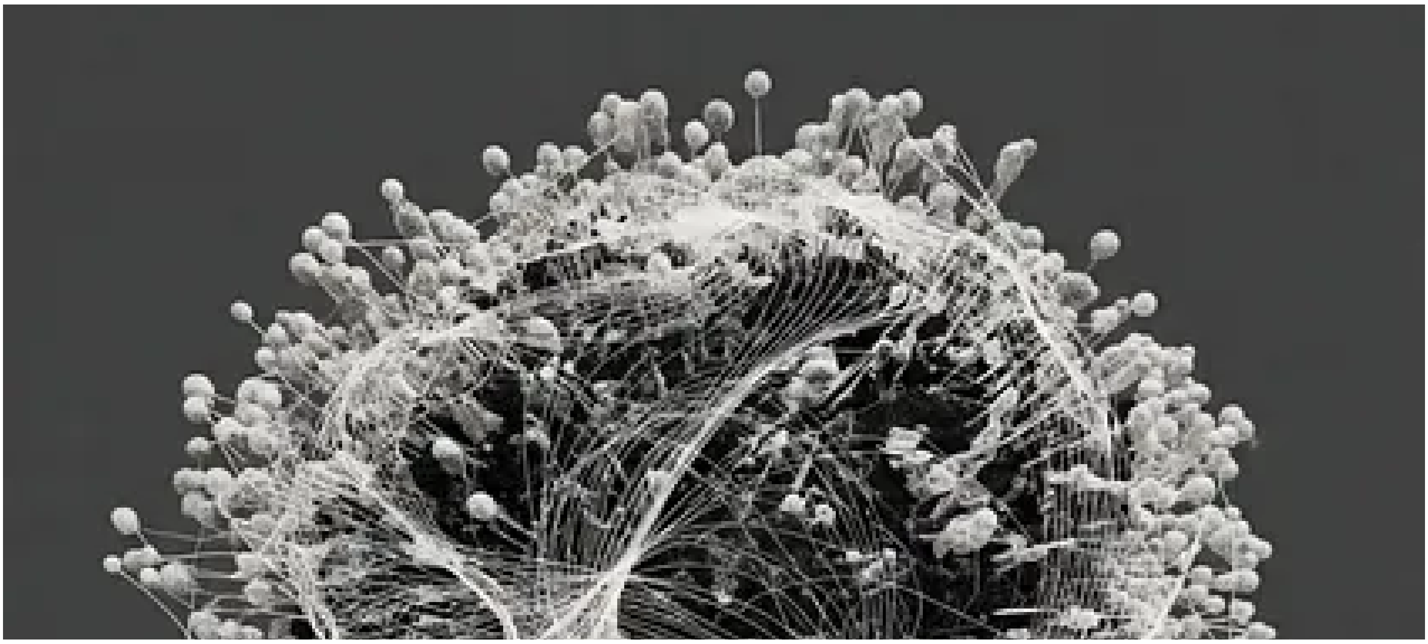

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Further Reading

Finite Tractus: The Hidden Geometry of Language and Thought

Pairwise Phase Space Embedding in Transformer Architecture

Introducing the Takens-Based Transformer

JPEG Compression of LLM Input Embeddings

Introducing the Attralucian Essays

Semantic Uncertainty

About the author

Earlier works from a career in Biomedical Engineering

Note: most articles are posted on both Substack and Medium so you can choose your preferred format for reading.

Omne quod est, finitum est; tantum per mensuram cognosci potest

Everything that exists is finite; it can only be known by measure

> *kevinhaylett.substack.com* | *geofinitism.com*
>
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Keywords: Finite Symbolic Mechanics, Nonlinear Dynamics, Takens Theorem, philosophy, Mathematics, Finite Mechanics

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